

# Maximal unstable uniformity is independent of the complement

A candidate full negative resolution of Remark 39(b) in arXiv:2411.06209

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27 August 2026

**Status.** Candidate full negative solution, likely valid, pending expert review. The theorem below proves that the maximal unstable uniformity dimension is independent of the complementary space in every dimension, thereby disproving the existential conjecture in Remark 39(b) of the source paper.

## 1 Source conjecture

Czornik, Kitzing, and Siegmund study the one-sided system

$$x(n+1) = A(n)x(n), \quad n \in \mathbb{N}, \quad (1)$$

where  $A(n) \in \text{GL}(\mathbb{R}^d)$  and both  $A$  and  $A^{-1}$  are bounded [1]. On PDF and printed page 20, Remark 39(b) conjectures:

“There exists  $A: \mathbb{N} \rightarrow \text{GL}(\mathbb{R}^d)$  such that the maximal uniformity dimension  $u_2$  depends on the choice of  $L_2$ .”

Here  $L_2$  ranges over complements of the fixed stable space  $L_1$ .

## 2 Definitions and notation

Write

$$\Phi(0) = I, \quad \Phi(n) = A(n-1) \cdots A(0) \quad (n \geq 1),$$

so that the solution with initial value  $x_0$  is  $x(n, x_0) = \Phi(n)x_0$ . A splitting is  $\mathbb{R}^d = L_1 \oplus L_2$ . For a subspace  $L$ , let  $\mathcal{G}_j(L)$  be its  $j$ -dimensional subspaces.

Following the source, the system has a *dichotomy with uniformity dimensions*  $(j_1, j_2)$  on  $(L_1, L_2)$  if there is a single  $\alpha > 0$  such that

$$\|\Phi(n)u\| \leq C_U e^{-\alpha(n-m)} \|\Phi(m)u\|, \quad u \in U, \quad U \in \mathcal{G}_{j_1}(L_1), \quad (2)$$

$$\|\Phi(n)v\| \geq C_V e^{\alpha(n-m)} \|\Phi(m)v\|, \quad v \in V, \quad V \in \mathcal{G}_{j_2}(L_2), \quad (3)$$

for all  $n \geq m \geq 0$ , where the positive constants may depend on the subspace. Refining a uniformity subspace preserves the estimate and its exponent. Thus a dichotomy is uniform on every one-dimensional subspace of both splitting spaces [1, Definition 15 and Theorem 7].

The pair  $(u_1, u_2)$  is *maximal* if it is an available uniformity pair and every available pair  $(\ell_1, \ell_2)$  satisfies  $\ell_i \leq u_i$  [1, Definition 35].

$$\leq (1 + K^2 C') \|x(k, \pi_{L_2} v')\|, \quad k \in \mathbb{N}$$

and therefore

$$\|x(m, y(v'))\| \geq (1 + K^2 C')^{-1} \|x(m, v')\|, \quad m \in \mathbb{N}. \quad (21)$$

From the inequalities (20) and (21), we get

$$\|x(n, v')\| \geq e^{\alpha(n-m)} C'' (1 + K^2 C')^{-1} \|x(m, v')\|, \quad m, n \in \mathbb{N}, n \geq m \geq m_0.$$

The statement now follows from Lemma 36.  $\square$

**Theorem 38** (Maximal uniformity dimensions for one-sided time). *Suppose that  $T = \mathbb{N}$ . Let  $(L_1, L_2)$  be a splitting of  $\mathbb{R}^d$ . Suppose that System (1) has a dichotomy on  $(L_1, L_2)$  with maximal uniformity dimensions  $(u_1, u_2)$ . Denote by  $(u_1, u_2^\perp)$  the maximal uniformity dimensions of system (1) with respect to the splitting  $(L_1, L_1^\perp)$ . Then it holds that  $u_2 \leq u_2^\perp$ .*

*Proof.* This follows from Theorem 37, noting that for each  $V' \in \mathcal{G}_{u_2}(L_1^\perp)$ , we have  $\dim V_{L_1}[V'] = 0 \leq u_1$   $\square$

In the following remark we list those cases in which Theorem 37 yields a unique maximal uniformity dimension of a dichotomy, i.e. independent of the splitting, and we formulate a conjecture on the general dependence of the maximal uniformity dimensions on the subspace  $L_2$  which is complementary to  $L_1$ .

**Remark 39** (Dependence of maximal uniformity dimensions on splitting for one-sided time). *Suppose that system (1) with  $\mathbb{T} = \mathbb{N}$  has a dichotomy on a splitting  $(L_1, L_2)$ .*

(a) *The maximal uniformity dimensions do not depend on  $L_2$  in case  $\dim L_1 = 1$  or  $\dim L_1 = d - 1$ .*

(b) *Conjecture: There exists  $A: \mathbb{N} \rightarrow \text{GL}(\mathbb{R}^d)$  such that the maximal uniformity dimension  $u_2$  depends on the choice of  $L_2$ .*

Figure 1: Theorem 38 and Remark 39 in the source paper, page 20. Part (b) is the conjecture resolved below.

### 3 Proof intuition

An alternate complement is a graph over the original complement. If  $w = v + Tv$  lies in that graph, then  $v$  is in the old unstable space and  $Tv$  is in the stable space. The source proof tries to control  $T(V)$  by a full stable uniformity estimate, which leads to its extra dimension hypothesis. That is more than is needed.

Indeed, finitely many stable basis-line estimates give the absolute bound  $\|\Phi(n)s\| \leq Ke^{-\alpha n} \|s\|$  for every  $s \in L_1$ . Meanwhile the old unstable estimate gives  $\|\Phi(n)v\| \geq ce^{\alpha n} \|v\|$ . Hence the graph correction is at most a constant times  $e^{-2\alpha n}$  of the unstable component. It is therefore harmless after a finite time, and invertibility patches the finite initial interval.

### 4 Complement-transfer theorem

**Lemma 1** (Finite-time patch). *Let  $U \subset \mathbb{R}^d$  be fixed. Suppose that for some  $C > 0$ ,  $\alpha > 0$ , and  $r_0 \in \mathbb{N}$ ,*

$$\|\Phi(n)u\| \geq Ce^{\alpha(n-m)} \|\Phi(m)u\| \quad (n \geq m \geq r_0, u \in U).$$

*Then the same estimate holds for all  $n \geq m \geq 0$  after replacing  $C$  by a possibly smaller positive constant.*

*Proof.* Every  $\Phi(r)$  is invertible. For each  $m < r_0$ , compactness of the unit sphere of  $U$  gives

$$q_m := \inf_{0 \neq u \in U} \frac{\|\Phi(r_0)u\|}{\|\Phi(m)u\|} > 0.$$

For  $n \geq r_0 > m$ , apply the assumed estimate from  $r_0$  to  $n$  and insert  $q_m$ ; this gives the desired form with constant  $Cq_me^{-\alpha(r_0-m)}$ . The remaining pairs  $m \leq n < r_0$  are finite in number, and the corresponding norm ratios again have positive minima on the unit sphere of  $U$ . Taking the minimum of these finitely many constants proves the claim. This is also the mechanism of Lemma 36 in the source.  $\square$

**Theorem 1** (Uniformity dimensions transfer between all complements). *Let  $A: \mathbb{N} \rightarrow \text{GL}(\mathbb{R}^d)$  satisfy the standing boundedness assumptions of (1), and let*

$$\mathbb{R}^d = L_1 \oplus L_2 = L_1 \oplus L'_2.$$

*If the system has a dichotomy with uniformity dimensions  $(j_1, j_2)$  on  $(L_1, L_2)$ , then it has a dichotomy with the same uniformity dimensions on  $(L_1, L'_2)$ . The same exponent  $\alpha$  can be used.*

*Proof.* The degenerate cases  $L_1 = \{0\}$  or  $L_1 = \mathbb{R}^d$  have a unique complement, so assume  $1 \leq k := \dim L_1 \leq d - 1$ . An admissible uniformity pair then has  $j_1, j_2 \geq 1$ .

Let  $e_1, \dots, e_k$  be an orthonormal basis of  $L_1$ . By refinement of the stable estimate, for each basis line there is  $C_i > 0$  such that, on setting  $m = 0$  in (2),

$$\|\Phi(n)e_i\| \leq C_i e^{-\alpha n} \quad (n \geq 0).$$

Consequently, for  $s = \sum_i a_i e_i$ ,

$$\|\Phi(n)s\| \leq e^{-\alpha n} \sum_i |a_i| C_i \leq K e^{-\alpha n} \|s\|, \quad K := \left( \sum_i C_i^2 \right)^{1/2}. \quad (4)$$

This is an absolute time-zero estimate on all of  $L_1$ ; it does not assert moving-time uniformity on  $L_1$ .

Fix an arbitrary  $V' \in \mathcal{G}_{j_2}(L'_2)$ . Let  $P: \mathbb{R}^d \rightarrow L_2$  be the projection along  $L_1$ . Since  $V' \cap L_1 = \{0\}$ ,  $P|_{V'}$  is injective, and

$$V := P[V'] \in \mathcal{G}_{j_2}(L_2).$$

Thus  $V'$  is the graph of a linear map  $T: V \rightarrow L_1$ :

$$V' = \{v + Tv : v \in V\}.$$

Put  $M = \|T\|$ . By (3) on the fixed space  $V$ , there is  $c > 0$  such that

$$\|\Phi(n)v\| \geq c e^{\alpha(n-m)} \|\Phi(m)v\| \quad (n \geq m \geq 0, v \in V). \quad (5)$$

In particular, at  $m = 0$ ,  $\|\Phi(r)v\| \geq c e^{\alpha r} \|v\|$ . Combining this with (4), for every  $r \geq 0$  and  $v \in V$ ,

$$\|\Phi(r)Tv\| \leq K M e^{-\alpha r} \|v\| \leq D e^{-2\alpha r} \|\Phi(r)v\|, \quad D := \frac{KM}{c}. \quad (6)$$

Choose  $r_0$  so that  $D e^{-2\alpha r_0} \leq \frac{1}{2}$ . For  $w = v + Tv \in V'$  and  $n \geq m \geq r_0$ , equations (5) and (6) give

$$\begin{aligned} \|\Phi(n)w\| &\geq \|\Phi(n)v\| - \|\Phi(n)Tv\| \geq \frac{1}{2} \|\Phi(n)v\|, \\ \|\Phi(m)w\| &\leq \|\Phi(m)v\| + \|\Phi(m)Tv\| \leq (1 + D) \|\Phi(m)v\|. \end{aligned}$$

Therefore

$$\|\Phi(n)w\| \geq \frac{c}{2(1+D)} e^{\alpha(n-m)} \|\Phi(m)w\| \quad (n \geq m \geq r_0, w \in V').$$

Lemma 1 extends this to all  $n \geq m \geq 0$ . Hence the unstable estimate holds uniformly on  $V'$  with exponent  $\alpha$ . Since  $V'$  was an arbitrary member of  $\mathcal{G}_{j_2}(L'_2)$ , all required unstable estimates hold on the new complement. The stable estimates on  $\mathcal{G}_{j_1}(L_1)$  are unchanged. Thus  $(j_1, j_2)$  transfers to  $(L_1, L'_2)$  with the same exponent.  $\square$

**Corollary 1** (Full negative answer to Remark 39(b)). *For one-sided time and fixed  $L_1$ , the maximal unstable uniformity dimension  $u_2$  is independent of the choice of complement  $L_2$ . Hence the system conjectured in Remark 39(b) of [1] does not exist.*

*Proof.* Let  $u_2$  and  $u'_2$  be the maximal unstable uniformity dimensions for two complements  $L_2$  and  $L'_2$ . Theorem 1 transfers the maximal pair for  $(L_1, L_2)$  to  $(L_1, L'_2)$ , so maximality gives  $u_2 \leq u'_2$ . Applying the same theorem in the reverse direction gives  $u'_2 \leq u_2$ . Thus  $u_2 = u'_2$ .  $\square$

## 5 Why the source hypothesis is unnecessary

Theorem 37 of [1] proves  $u_2 \leq u'_2$  under the additional condition that the stable projection of every target  $u_2$ -plane has dimension at most  $u_1$ . That condition supplies a moving-time stable estimate on the entire projected plane. The proof above replaces it with (4), obtained from finitely many basis lines and used only at absolute time zero. This is sufficient because the stable and unstable absolute-time bounds separate at the rate  $e^{-2\alpha r}$ . No dimension restriction on the stable projection remains.

## 6 Verification, novelty, and limitations

The proof uses only finite-dimensional linear algebra, the definitions and refinement observation in the source, and the explicitly proved finite-time patch. The constants  $c$ ,  $D$ , and  $r_0$  may depend on the chosen target plane  $V'$ , exactly as the source definition permits; the exponent  $\alpha$  does not. Degenerate splittings have a unique complement.

A bounded novelty check on 27 August 2026 searched the run’s lightweight indexes and live arXiv/web results by the exact arXiv id, paper title, exact conjecture language, authors, and the phrases “maximal uniformity dimension”, “dependence on splitting”, and “choice of  $L_2$ ”. It found the source paper and a repository mirror, but no later claimed resolution. This is not an exhaustive novelty guarantee.

There is no computational component. Human review should focus on the common exponent under refinement, the graph parametrization of an arbitrary target  $j_2$ -plane, and the fact that only the absolute stable estimate (4) is used.

## References

- [1] A. Czornik, K. Kitzing, and S. Siegmund, *Dichotomies uniform on subspaces and formulas for dichotomy spectra*, arXiv:2411.06209, latest source/PDF inspected 27 August 2026.